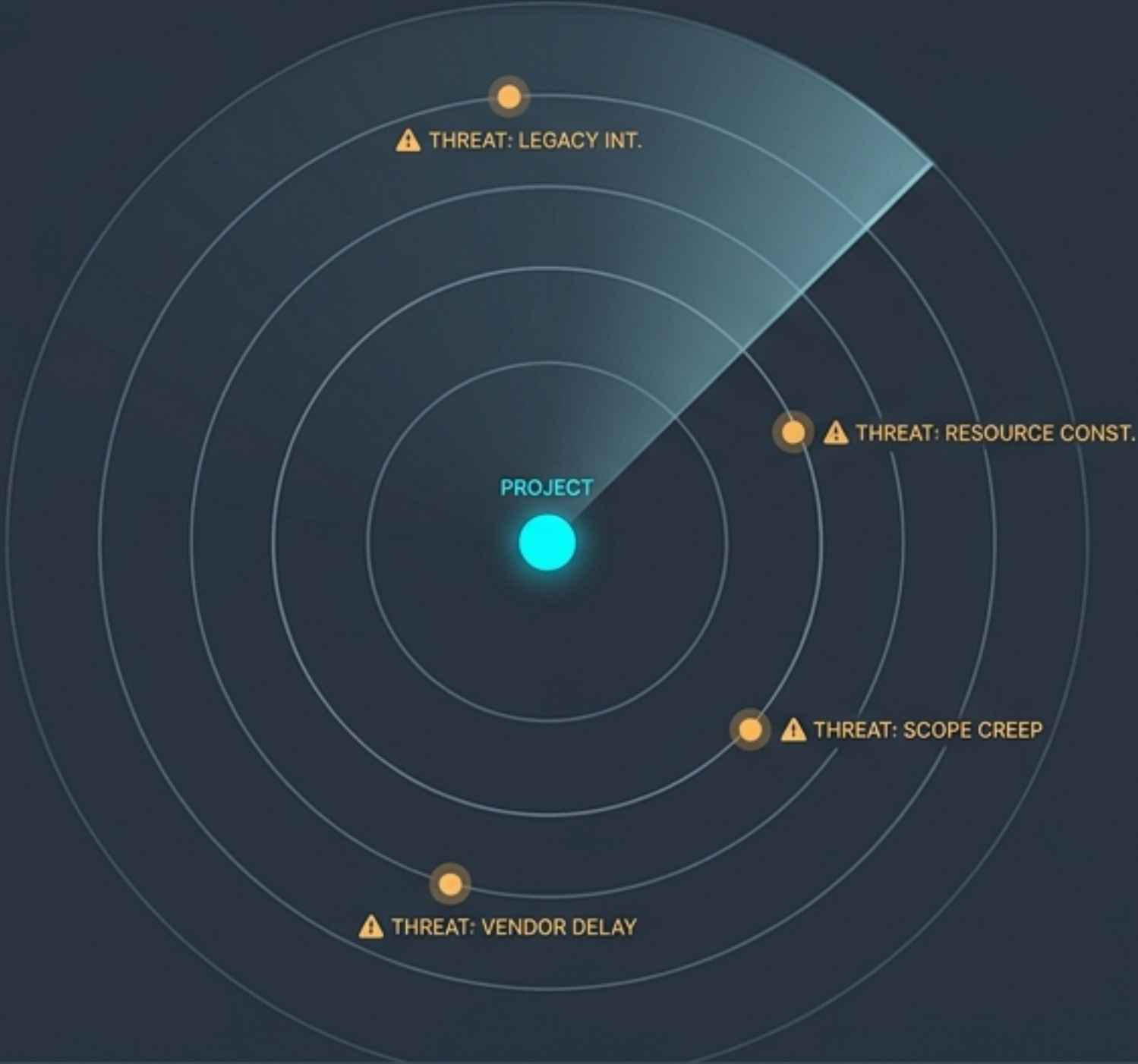


Engineering Predictability: The Software Risk Playbook

Transforming project uncertainty into manageable data using the Pressman Framework.



RISK ASSESSMENT MATRIX (PRESSMAN)

		IMPACT		
		High	Medium	Low
PROBABILITY	High	⚠️ CRITICAL RISK: IMMEDIATE MITIGATION		
	Medium		🔍 MONITOR: PRGACTIVE CONTROL	
	Low			✅ ACCEPTABLE: ROUTINE MONITORING

PROJECT LIFECYCLE: RISK & CONTROL POINTS



The Anatomy of a Project Risk



ISSUE

What can go wrong?



PROBABILITY

What is the likelihood?



IMPACT

What will the damage be?



MITIGATION

What can we do about it?

Risk management is the systematic process of quantifying these four variables before they impact the project lifecycle.

The Engineering Paradigm: Reactive vs. Proactive

REACTIVE

The Firefighting Trap

- **Reacts to occurrences:** Waits for the risk to strike.
- **Fix on Failure:** Resources are scrambled and applied only during the strike.
- **Crisis Management:** Scrambling to recover from events leading to project failure.
- **Mitigation as an afterthought:** Planning for additional resources merely in anticipation of failure.

PROACTIVE

The Radar Approach

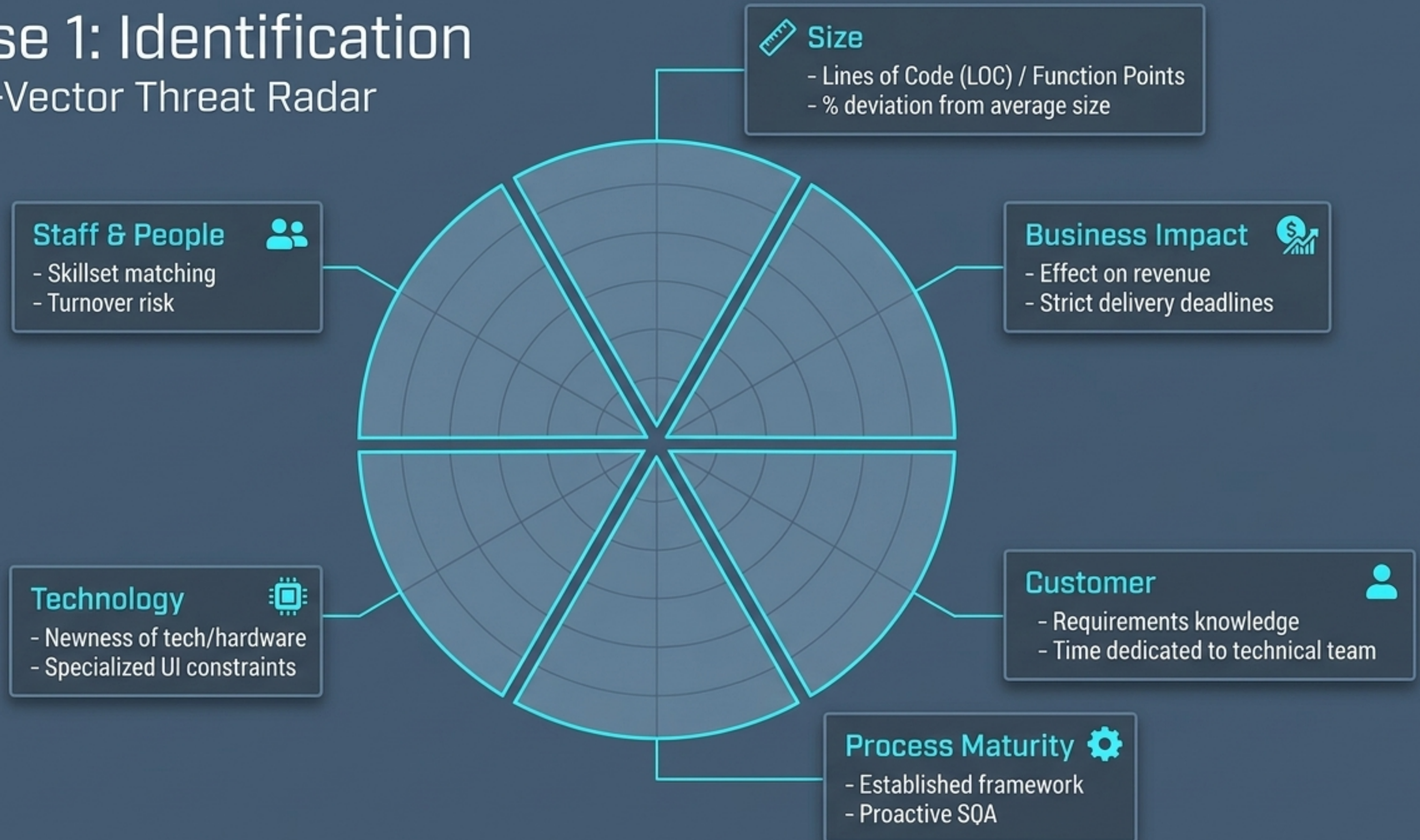
- **Formal Risk Analysis:** Performed systematically before coding begins.
- **Root Cause Correction:** Fixing the foundational issues, not just the symptoms.
- **Beyond Software Bounds:** Examining risk sources across the entire business and customer landscape.
- **Managing Change:** Developing organizational muscle to handle shifting scopes smoothly.

The 7 Principles of Risk Avoidance



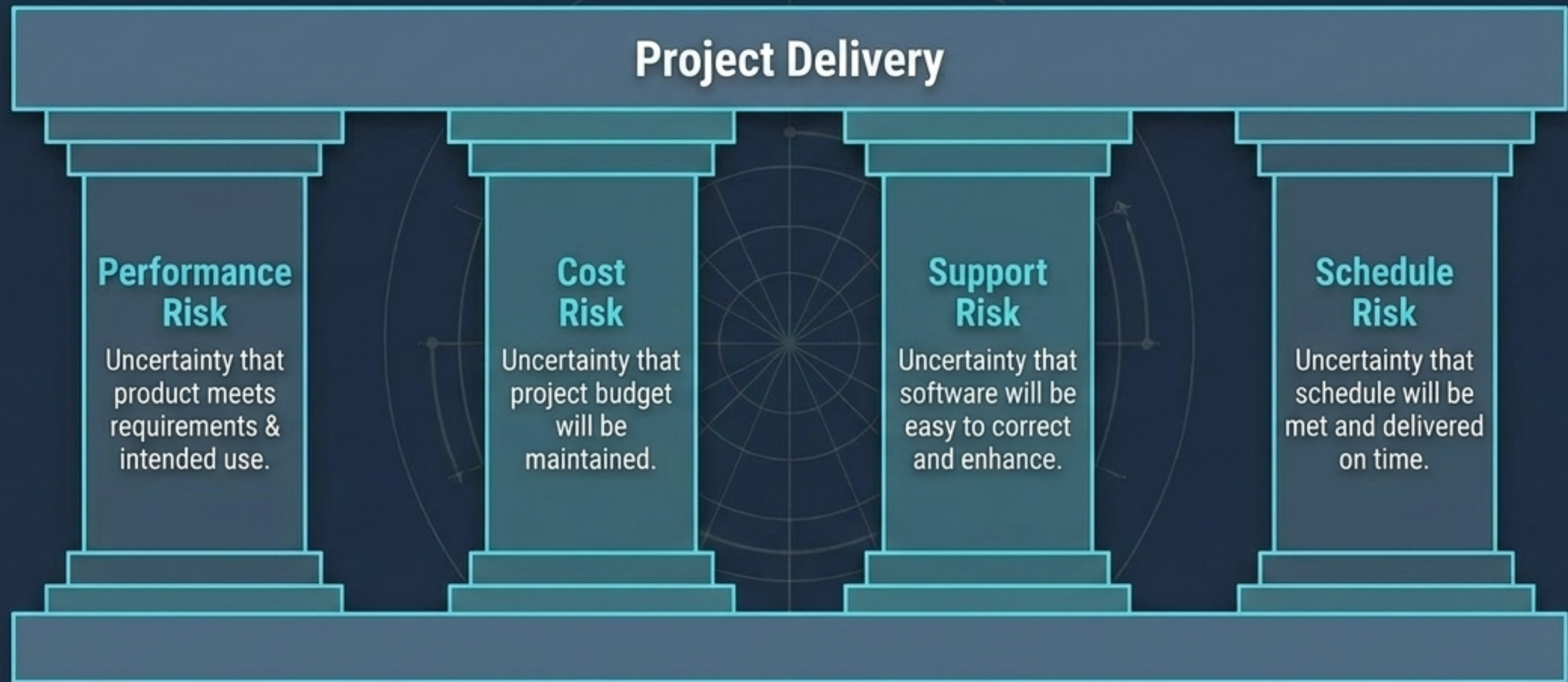
Phase 1: Identification

The 6-Vector Threat Radar



Phase 2: Assessment

The Structural Integrity of a Project



Assessment relies on answering critical structural questions:
Are requirements stable? Is the scope locked? Do we have top management support?

Phase 3: Estimation (Calculating Risk Exposure)



$$RE = P \times C$$

RE: Overall Risk Exposure

P: Probability of occurrence

C: Cost to the project if it occurs

Risks are sorted and prioritized by this calculated exposure weight, separating highly probable catastrophic threats from unlikely minor inconveniences.

Estimation in Practice: The Cost of Failed Reuse

Scenario: 60 reusable software components planned.

Risk: Only 70% will integrate successfully. The rest require custom development.

Probability (P)

80%

Likely

Deficit

18 components
must be built from
scratch.

Unit Cost

100 LOC/component
@ \$14.00/LOC

Cost/Impact (C)

$18 \times 100 \times \$14$
 $= \$25,200$

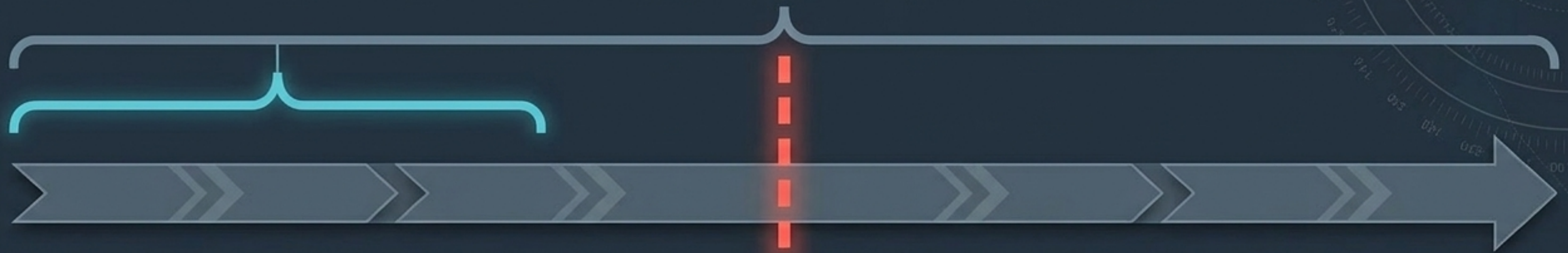
$$\text{Risk Exposure (RE)} = 0.80 \times \$25,200 \approx \$20,200$$

This risk carries a literal **\$20,200 weight** on the project ledger. This number determines how much budget should be allocated to mitigate it.

Phase 4: Execution

The RMMM Framework

Risk Event Trigger



MITIGATION

- How can we avoid the risk entirely?
- **Action:** Proactive steps taken before the project begins or during early development to remove the threat vector.

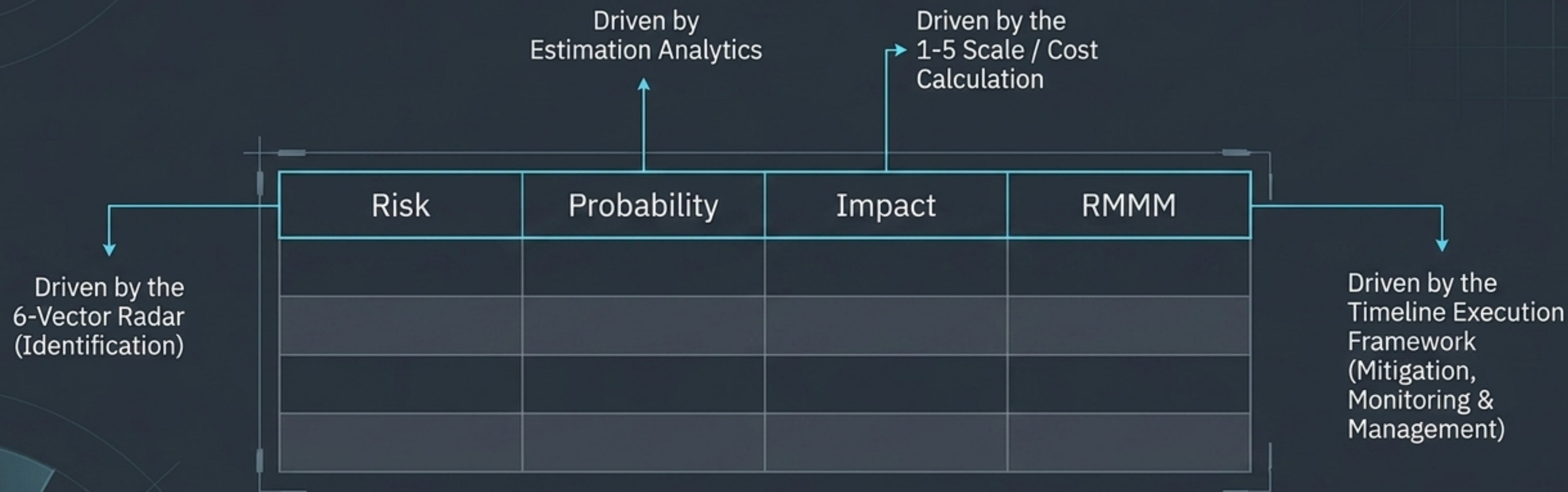
MONITORING

- What factors can we track to determine if the risk is becoming more/less likely?
- **Action:** Continuous data gathering (e.g., tracking milestone completion, turnover rates).

MANAGEMENT

- What contingency plans do we execute if the risk becomes a reality?
- **Action:** Executing the predefined Plan B to stabilize the project.

The Unified Framework: The Risk Table



Workflow Note: The table is continuously sorted by Probability and Impact (Risk Exposure), ensuring the team is always focused on the heaviest threats.

The Output: Formal Risk Documentation

dossier

Project: Embedded software for XYZ system

Risk type: Schedule risk

Priority (1 low ... 5 critical): 4

Risk factor: Project completion depends on hardware tests. Hardware delivery may be delayed.

Probability: 60%

Impact: Project delayed each day hardware is unavailable.

Monitoring approach: Scheduled milestone reviews with hardware group.

Contingency plan: Modify testing strategy using software simulation.

Estimated resources: 6 additional person-months beginning in July.

Predictability is not achieved by avoiding risks, but by documenting, estimating, and managing them with unyielding engineering discipline.